

Worksheet - 1

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Aim:

To understand the usage of fundamental Python libraries for Machine Learning and perform comprehensive data preprocessing on the Titanic dataset.

Objective:

To explore and use fundamental Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn for data analysis and visualization, and to perform comprehensive data preprocessing on the Titanic dataset by handling missing values, removing duplicate records, encoding categorical variables, detecting and treating outliers, and preparing the dataset for Machine Learning model implementation.

Input/ Apparatus Used:

PyCharm

Part A: Implementation of Python Basic Libraries

import math

```
print("square Root:", math.sqrt(25))
print("Factorial:", math.factorial(5))
print("Power:", math.pow(2,3))
print("ceiling:", math.ceil(4.3))
print("floor:", math.floor(4.7))
print("Pi value:", math.pi)
print("Euler's number:", math.e)
```

import numpy as np

```
arr = np.array([1,2,3,4,5])
```

```
print("Array:", arr)
print("Mean:", np.mean(arr))
print("Sum:", np.sum(arr))
print("Max:", np.max(arr))
```

```
matrix = np.array([[1,2,3],[4,5,6],[7,8,9]])
print("Matrix:\n", matrix)
print("Transpose:\n", matrix.T)
```

#Matplotlib

```
import matplotlib.pyplot as plt
```

```
x = [1,2,3,4,5]
y = [10,20,30,35,50]
```

```
plt.plot(x,y)
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Simple Line Graph")
plt.show()
```

```
data = [10,20,30,40,50]
```

```
#seaborn
import seaborn as sns
import matplotlib.pyplot as plt
```

```
sns.histplot(data)
plt.title("Seaborn Histogram")
plt.show()
```

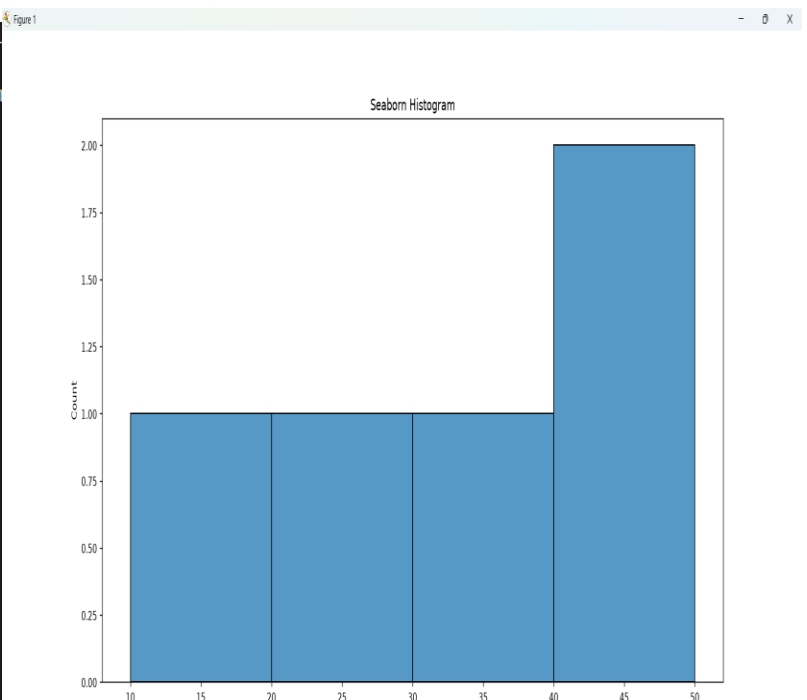
```
# Scipy
from scipy import stats
```

```
data = [5,10,15,20,25]
```

```
print("mean:", stats.tmean(data))
print("median:", stats.scoreatpercentile(data, 50))
print("mode:", stats.mode(data))
```

Output:

```
PS C:\Users\A\OneDrive\Desktop\machine learning> .venv\Scripts\Activate.ps1
(.venv) PS C:\Users\A\OneDrive\Desktop\machine learning> python firsrex.py
square Root: 5.0
Factorial: 120
Power: 8.0
ceiling: 5
floor: 4
Pi value: 3.141592653589793
Euler's number: 2.718281828459045
Array: [1 2 3 4 5]
Mean: 3.0
Sum: 15
Max: 5
Matrix:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
Transpose:
[[1 4 7]
 [2 5 8]
 [3 6 9]]
```



Part B: Data Preprocessing on Titanic Dataset

```
import seaborn as sns
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.preprocessing import StandardScaler

df = sns.load_dataset('titanic')

print(df.head())

print(df.isnull().sum())

df['age'].fillna(df['age'].median(), inplace=True)
df['embarked'].fillna(df['embarked'].mode()[0], inplace=True)
df.drop(columns=['deck'], inplace=True)

sns.boxplot(x=df['fare'])
plt.show()

Q1 = df['fare'].quantile(0.25)
Q3 = df['fare'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df['fare'] >= lower) & (df['fare'] <= upper)]

scaler = StandardScaler()
df[['age', 'fare']] = scaler.fit_transform(df[['age', 'fare']])

df.drop_duplicates(inplace=True)
print(df.info())
print(df.head())
```

Output:

```
(.venv) PS C:\Users\A\OneDrive\Desktop\machine learning> & "c:\Users\A\OneDrive\Desktop\machine learning\.venv\Scripts\python.exe  
irstexp2.py"  
   survived  pclass    sex   age  sibsp  parch    fare embarked  class  who  adult_male  deck  embark_town  alive  alone  
0          0        3   male  22.0    1     0   7.2500      S  Third   man         True   NaN  Southampton   no  False  
1          1        1  female  38.0    1     0  71.2833      C  First  woman        False    C    Cherbourg   yes  False  
2          1        3  female  26.0    0     0   7.9250      S  Third  woman        False   NaN  Southampton   yes  True  
3          1        1  female  35.0    1     0  53.1000      S  First  woman        False    C    Southampton   yes  False  
4          0        3   male  35.0    0     0   8.0500      S  Third   man         True   NaN  Southampton   no   True  
survived      0  
pclass        0  
sex           0  
age          177  
sibsp         0  
parch         0  
fare          0  
embarked      2  
class         0  
who           0  
adult_male    0  
deck         688  
embark_town   2  
alive         0  
alone         0  
dtype: int64
```

Learning Outcomes:

1. Use Python libraries such as **NumPy, Pandas, Matplotlib, and Seaborn**.
2. Load and analyze the **Titanic dataset**.
3. Handle missing and duplicate data.
4. Perform data cleaning and preprocessing.
5. Prepare the dataset for machine learning models.

